Experimental Setup

Diffusion, Score Matching and Flow Matching on MNIST

Continuous Normalizing Flows

Data space

$$x = (x_1, \dots, x_d) \in \mathbb{R}^d$$

Probability density path

$$\boxed{p : [0,1] \times \mathbb{R}^d \to \mathbb{R}_{>0}} \quad \text{where} \int_{\mathbb{R}^d} p_t(x)\, dx = 1.$$

Time-dependent vector field

$$\boxed{v : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d}$$

A vector field defines a flow ϕ_t through

$$\boxed{\frac{d}{dt}\phi_t(x) = v_t(\phi_t(x)), \qquad \phi_0(x) = x.}$$

1. Unified Time Convention

All five methods use the same Gaussian conditional path

$$\boxed{x_t = \alpha(t)x_0 + \sigma(t)x_1}$$

where

$$x_0 \sim \mathcal{N}(0, I), \qquad x_1 \sim p_{\mathrm{data}}.$$

The time interval is

$$\boxed{t \in [0,\ 1]}$$

with

$$t = 0: \qquad \alpha(0) \approx 1, \qquad \sigma(0) \approx 0,$$

$$t = 1: \qquad \alpha(1) \approx 0, \qquad \sigma(1) \approx 1.$$

2. Noise-to-Data Probability Path

The conditional path is

$$x_t = \alpha(t)x_0 + \sigma(t)x_1.$$

Here

$$\underbrace{x_0}_{\text{Gaussian noise}} \sim \mathcal{N}(0, I), \qquad \underbrace{x_1}_{\text{data}} \sim p_{\mathrm{data}}.$$

At the two boundaries:

$$t \approx 0: \qquad x_t \approx x_0 \sim \mathcal{N}(0, I),$$

$$t \approx 1: \qquad x_t \approx x_1 \sim p_{\mathrm{data}}.$$

Thus the generative process is

$$\boxed{\mathcal{N}(0, I) \longrightarrow p_{\mathrm{data}}}$$

3. Five Methods

The implementation contains five methods based on the same probability-path framework.

Method	Probability path	Network target
FM-OT	Linear / OT	velocity v_θ
FM-Diffusion	VP	velocity v_θ
DDPM	VP	noise ϵ_θ
Score	VP	score s_θ
Score Flow	VP	score s_θ

The comparison is performed using

$$\boxed{\text{same UNet} + \text{same time convention} + \text{same sampling interface}}$$

Probability path

$$\boxed{x_t = \alpha(t)x_0 + \sigma(t)x_1}$$

with

$$\boxed{\alpha(t) = 1 - t, \qquad \sigma(t) = t}$$

Therefore,

$$\boxed{x_t = (1 - t)x_0 + tx_1}$$

Conditional velocity

$$\dot{\alpha}(t) = -1, \qquad \dot{\sigma}(t) = 1$$

$$\boxed{u_t = \dot{\alpha}(t)x_0 + \dot{\sigma}(t)x_1}$$

$$\boxed{u_t = x_1 - x_0}$$

$$\boxed{\frac{\partial u_t}{\partial t} = 0}$$

$$\boxed{x_0 \sim \mathcal{N}(0, I) \xrightarrow{\text{Linear / OT path}} x_1 \sim p_{\text{data}}}$$

5. Variance-Preserving Probability Path

Probability path

$$\boxed{x_t = \alpha(t)x_0 + \sigma(t)x_1}$$

with

$$\boxed{\alpha(t)^2 + \sigma(t)^2 = 1}$$

and

$$\alpha : 1 \to 0, \qquad \sigma : 0 \to 1.$$

Noise rate

$$\boxed{\beta(t) = -2\frac{d}{dt}\log\alpha(t)}$$

Used by:

FM-Diffusion
DDPM
Score
Score Flow

$$\boxed{x_0 \sim \mathcal{N}(0, I) \quad \xrightarrow{\text{VP path}} \quad x_1 \sim p_{\mathrm{data}}}$$

6. Conditional Vector Field

Given

$$x_t = \alpha(t)x_0 + \sigma(t)x_1,$$

differentiate with respect to t:

$$\frac{dx_t}{dt} = \dot{\alpha}(t)x_0 + \dot{\sigma}(t)x_1.$$

Therefore,

$$\boxed{\begin{aligned} u_t(x|x_0, x_1) &= \dot{\alpha}(t)x_0 + \dot{\sigma}(t)x_1, \\ v_t(x) &= \mathbb{E}\left[u_t(x|x_0, x_1) \mid x_t = x\right]. \end{aligned}}$$

Flow Matching trains a neural network to approximate

$$\boxed{v_\theta(x, t) \approx v_t(x).}$$

7. Probability-Flow ODE

Sampling dynamics

$$\boxed{\frac{dx}{dt} = v_\theta(x, t)}$$

Initial state:

$$\boxed{x(0) \sim \mathcal{N}(0, I)}$$

Forward integration:

$$\boxed{0 \longrightarrow 1}$$

Generated sample

At the final time,

$$\boxed{x(1) \approx x_{\text{data}}}$$

All methods use the same interface:

$$\boxed{v_\theta(x, t) \to \text{ODE solver}}$$

$$\text{Noise} \quad \xrightarrow{v_\theta} \quad \text{Data}$$

8. Flow Matching Velocity

For Flow Matching, the network directly predicts velocity:

$$\boxed{v_\theta(x_t, t) \approx u_t(x_t | x_0, x_1).}$$

The target is

$$\boxed{u_t = \dot{\alpha}(t) x_0 + \dot{\sigma}(t) x_1.}$$

For FM-OT,

$$\boxed{u_t = x_1 - x_0.}$$

For FM-Diffusion using the VP schedule.

$$\boxed{u_t = \dot{\alpha}(t) x_0 + \dot{\sigma}(t) x_1}$$

Therefore the two Flow Matching methods differ mainly in their choice of probability path.

9. Score and Noise Parameterization

For the Gaussian conditional distribution,

$$\sigma(t)x_1 + \alpha(t)x_0 = x_t \sim p_t(x_t|x_1) = \mathcal{N}\left(x_t; \sigma(t)x_1, \alpha(t)^2 I\right).$$

Its conditional score is

$$\boxed{\nabla_{x_t} \log p_t(x_t|x_1) = -\frac{x_t - \sigma(t)x_1}{\alpha(t)^2}.}$$

The conditional score becomes

$$s_t(x_t|x_1) - \frac{x_t - \sigma(t)x_1}{\alpha(t)^2} = -\frac{\alpha(t)x_0}{\alpha(t)^2} = \boxed{-\frac{x_0}{\alpha(t)}}.$$

Therefore,

$$\boxed{s_\theta(x,t) = -\frac{\epsilon_\theta(x,t)}{\alpha(t)}}$$

10. Score $\rightarrow$ Probability-Flow Velocity

For the SDE

$$dx = f(x,t)dt + g(t)dW_t,$$

the corresponding Probability-Flow ODE has velocity

$$\boxed{v(x,t) = f(x,t) - \frac{1}{2}g(t)^2 s(x,t).}$$

The score model therefore does not need to directly predict the ODE velocity.
Instead,

$$\boxed{s_\theta(x,t) \quad \Longrightarrow \quad v_\theta(x,t)}$$

through the Probability-Flow transformation.
Consequently, DDPM, Score and Score Flow can all be sampled using the same ODE interface as Flow Matching.

11. Unified Training Objective

All methods use the general objective

$$\boxed{\mathcal{L}(\theta) = \mathbb{E}\left[w(t)\,\|\hat{y}_\theta(x_t, t) - y^\star\|^2\right].}$$

Method	Target $y^\star$	Weight $w(t)$
FM-OT	$\dot{\alpha}x_0 + \dot{\sigma}x_1$	1
FM-Diffusion	$\dot{\alpha}x_0 + \dot{\sigma}x_1$	1
DDPM	x_0	1
Score	$-\dfrac{x_0}{\alpha(t)}$	$\alpha(t)^2$
Score Flow	$-\dfrac{x_0}{\alpha(t)}$	$\beta(t)$

12. Score Matching Loss

For the Score method,

$$y^{\star} = -\frac{x_0}{\alpha(t)}, w(t) = \alpha(t)^2.$$

Therefore,

$$\mathcal{L}_{\text{score}} = \mathbb{E}\left[\alpha(t)^2 \left\| s_\theta(x_t, t) + \frac{x_0}{\alpha(t)} \right\|^2\right].$$

Equivalently,

$$\boxed{\mathcal{L}_{\text{score}} = \mathbb{E}\left[\left\|\alpha(t) s_\theta(x_t, t) + x_0\right\|^2\right].}$$

The weighting compensates for the $1/\alpha(t)$ scale of the conditional score.

13. Score Flow Loss

Score Flow uses the same score target

$$\boxed{s_t^\star = -\frac{x_0}{\alpha(t)}}$$

but uses the likelihood-related weighting

$$\boxed{w(t) = \beta(t).}$$

Therefore,

$$\boxed{\mathcal{L}_{\text{ScoreFlow}} = \mathbb{E}\left[\beta(t)\left\|s_\theta(x_t, t) + \frac{x_0}{\alpha(t)}\right\|^2\right].}$$

The distinction between the two score-based objectives is

$$\boxed{\text{Score: } w(t) = \alpha(t)^2}$$

14. Metrics: NLL

Likelihood is evaluated using the Probability-Flow ODE.
The log-density evolves according to

$$\boxed{\frac{d \log p_t(x_t)}{dt} = -\nabla_x \cdot v_\theta(x_t, t).}$$

Therefore,

$$\boxed{\log p_{1-0}(x) = \log p_0(x) - \int_0^{1-0} \nabla_x \cdot v_\theta(x_t, t)\, dt.}$$

The divergence is estimated using Hutchinson's estimator:

$$\boxed{\nabla_x \cdot v_\theta \approx \epsilon^\top \frac{\partial v_\theta}{\partial x}\epsilon, \quad \epsilon_i \in \{-1, +1\}.}$$

The resulting likelihood is reported as bits per dimension (BPD).

Fréchet Inception Distance

$$\boxed{\mathrm{FID} = \|\mu_r - \mu_g\|_2^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right).}$$

The implementation evaluates generated and real MNIST images in a feature space.

Number of Function Evaluations

$$\boxed{\mathrm{NFE} = \text{number of calls to } v_\theta(x,t)}$$

required by the adaptive ODE solver.
Thus:

$$\boxed{\begin{array}{ll} \text{NLL} & \rightarrow \text{likelihood} \\ \text{FID} & \rightarrow \text{sample quality} \\ \text{NFE} & \rightarrow \text{sampling efficiency} \end{array}}$$

16. Experimental Results

MNIST-CNN Features

Method	NLL $\downarrow$	FID $\downarrow$	NFE $\downarrow$
DDPM	2.265	**89.627**	182
Score	2.176	130.730	155
Score Flow	2.678	309.768	164
FM-Diff.	2.607	104.448	173
FM-OT	**1.660**	96.205	**122**

Inception-v3 Features

Method	NLL $\downarrow$	FID $\downarrow$	NFE $\downarrow$
DDPM	2.265	136.674	182
Score	2.176	121.454	155
Score Flow	2.678	184.895	164
FM-Diff.	2.607	129.497	173
FM-OT	**1.660**	**119.785**	**122**

Key Finding

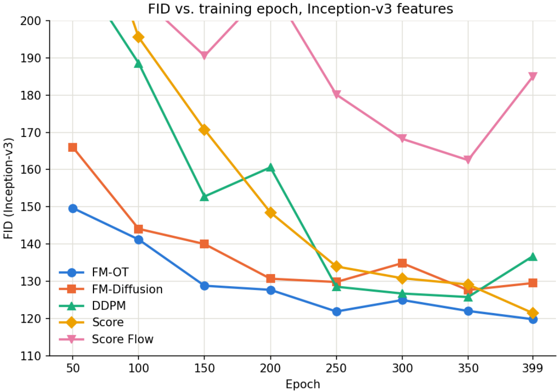
FM-OT achieves the lowest NLL and NFE in both feature spaces. The linear OT path produces a simpler velocity field, leading to more efficient ODE integration.

Epoch = 350 Samples = 2000

17. Experimental Results: FID vs. Epoch

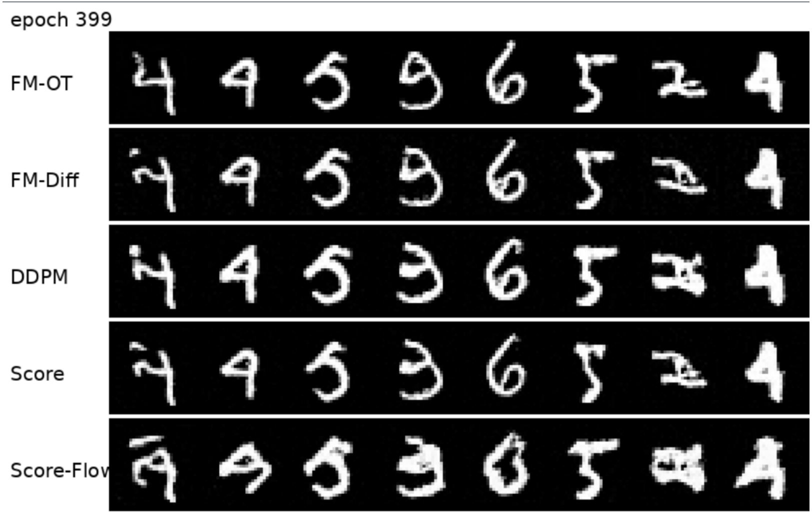
MNIST-CNN Features

Inception-v3 Features



FID $\downarrow$ vs. Training Epoch

18. Qualitative Results: Generated MNIST Samples



19. Main Experimental Findings

- **FM-OT achieves the lowest NLL**

$$\boxed{\mathrm{NLL}_{\mathrm{FM-OT}} = 1.660}$$

- **FM-OT requires the fewest function evaluations**

$$\boxed{\mathrm{NFE}_{\mathrm{FM-OT}} = 122}$$

  compared with

$$\mathrm{NFE}_{\mathrm{DDPM}} = 182.$$

- The linear OT path has a particularly simple conditional velocity:

$$\boxed{u_t = x_1 - x_0.}$$

- The same Probability-Flow ODE interface allows all five methods to be evaluated under the same sampling procedure.

20. Complete Implementation Pipeline

$$\boxed{x_0 \sim \mathcal{N}(0, I), \qquad x_1 \sim p_{\mathrm{data}}}$$

$$\Downarrow$$

$$\boxed{x_t = \alpha(t)x_0 + \sigma(t)x_1}$$

$$\Downarrow$$

$$\boxed{\mathrm{UNet}(x_t, t) \to \hat{y}_\theta}$$

where

$$\hat{y}_\theta \in \{v_\theta,\ \epsilon_\theta,\ s_\theta\}.$$